

NP Lemma: $X \sim f$ Test- $H_0: f = f_0$ vs. $H_1: f = f_1$
known known
 $P_0, P_1 = \text{Prob. with density } f_0, f_1$

1. For $0 < \alpha < 1 \exists$ a test given by $\phi^*(x) = \begin{cases} 1 & \text{if } f_1(x) > K f_0(x) \\ \gamma & \text{if } f_1(x) = K f_0(x) \\ 0 & \text{if } f_1(x) < K f_0(x) \end{cases}$

where $K \geq 0, \gamma \in [0, 1]$

are determined from $E_0[\phi^*] = \alpha$, is a MP size- α test

2. Uniqueness: If ϕ_0 is a MP size- α test, then

$$\phi_0(x) = \begin{cases} 1 & \text{if } f_1(x) > K f_0(x) \\ 0 & \text{if } f_1(x) < K f_0(x) \end{cases} \quad \mu \text{ a.e.}$$

$$B = \{x: \phi^*(x) \neq \phi_0(x), f_1(x) \neq K f_0(x)\}$$

$$P_0(B) = 0, P_1(B) = 0$$

Ex. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U(0, \theta), \theta > 0$

Test $H_0: \theta = \theta_0$ vs. $H_1: \theta = \theta_1 (> \theta_0)$. MP Size- α Test.

$$\lambda(x_{(n)}) = \frac{f(x|\theta_1)}{f(x|\theta_0)} = \begin{cases} \left(\frac{\theta_0}{\theta_1}\right)^n & \text{if } 0 \leq x_{(n)} < \theta_0 \\ \infty & \text{if } \theta_0 \leq x_{(n)} \leq \theta_1 \end{cases}$$

$=K_1$ $=K_2$

$K = K_1$:

$$\{\lambda(x_{(n)}) > K_1\} = \{\theta_0 \leq x_{(n)} \leq \theta_1\}$$

$$\{\lambda(x_{(n)}) = K_1\}$$

$$\phi(x) = \begin{cases} 1 & \text{if } \overbrace{\theta_0 \leq x_{(n)} \leq \theta_1}^{\lambda(x_{(n)}) > K_1} \\ \gamma & \text{if } \lambda(x_{(n)}) = K_1 \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{E}_{\theta_0}(\phi) = \alpha \Rightarrow \underbrace{\mathbb{P}_{\theta_0}(\theta_0 < X_{(n)} \leq \theta_1)}_{=0} + \underbrace{\gamma \mathbb{P}_{\theta_0}(0 \leq X_{(n)} \leq \theta_0)}_{=1} = \alpha$$

$$\phi(x) = \begin{cases} 1 & \text{if } x_{(n)} \in (\theta_0, \theta_1) \\ \alpha & \text{if } x_{(n)} \in (0, \theta_0) \end{cases} \xrightarrow{\gamma = \alpha} \text{randomized}$$

Claim: $\phi^*(x) = \begin{cases} 1 & \text{if } x_{(n)} > c \\ 0 & \text{if } x_{(n)} < c \end{cases}$ s.t. $\mathbb{E}_{\theta_0}(\phi^*) = \alpha$
is a MP size- α test

T.S. $\mathbb{E}_{\theta_1} \phi \leq \mathbb{E}_{\theta_1} \phi^*$ for $\phi: \mathbb{E}_{\theta_0} \phi = \alpha$

$A = \{x: \phi^*(x) \neq \phi(x)\}$ If $x \in A$, $\phi^*(x) > \phi(x) \Rightarrow x_{(n)} \geq c$
 $\Rightarrow \underbrace{\lambda(x_{(n)})}_{= \frac{f_{\theta_1}(x)}{f_{\theta_0}(x)}} \geq \lambda(c) := K$ (say)

If $x \in A$, $\phi^*(x) < \phi(x) \Rightarrow f_{\theta_1}(x) \leq K f_{\theta_0}(x)$

$$(\phi^*(x) - \phi(x))(f_{\theta_1}(x) - K f_{\theta_0}(x)) \geq 0 \quad \forall x \in A$$

$$\int_{\mathbb{R}} (\phi^*(x) - \phi(x))(f_{\theta_1}(x) - K f_{\theta_0}(x)) dx \geq 0$$

$$\int_{\mathbb{R}} (\phi^*(x) - \phi(x)) f_{\theta_1}(x) dx \geq K \int_{\mathbb{R}} (\phi^*(x) - \phi(x)) f_{\theta_0}(x) dx$$

$$\Rightarrow \mathbb{E}_{\theta_1} \phi^* - \mathbb{E}_{\theta_1} \phi \geq K (\underbrace{\mathbb{E}_{\theta_0} \phi^*}_{=\alpha} - \underbrace{\mathbb{E}_{\theta_0} \phi}_{=\alpha}) = 0$$

$$\Rightarrow \mathbb{E}_{\theta_1} \phi^* \geq \mathbb{E}_{\theta_1} \phi$$

$$P_{\theta_0}(X_{(n)} > c) = \alpha$$

$$c = (1-\alpha)^{1/n} \theta_0$$

$$\phi^*(x) = \begin{cases} 1 & \text{if } x_{(n)} > (1-\alpha)^{1/n} \theta_0 \\ 0 & \text{if } x_{(n)} < \end{cases}$$

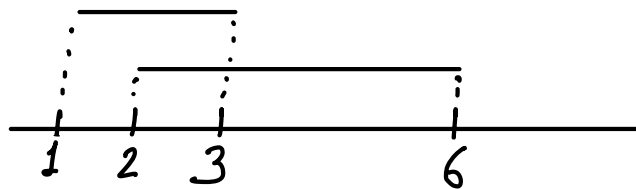
Non-uniqueness of MP test:

$$X \sim f \quad H_0: f = \underbrace{U(1,3)}_{f_0}, \quad H_1: f = \underbrace{U(2,6)}_{f_1}$$

$$f_0(x) = \frac{1}{2} \quad \text{if } 1 < x < 3$$

$$f_1(x) = \frac{1}{4} \quad \text{if } 2 < x < 6$$

Find MP size- α test, $\alpha = 0.05$.



$$\lambda(x) = \frac{f_1(x)}{f_0(x)} = \begin{cases} 0 & \text{if } 1 < x < 2 \\ 1/2 & \text{if } 2 < x < 3 \\ \infty & \text{if } 3 < x < 6 \end{cases}$$

$$K=0$$

$$\underbrace{P_0(2 < X < 3)}_{1/2} + \underbrace{P_0(3 < X < 6)}_0 + \gamma P_0(1 < X < 2) = 1/2 > \alpha$$

$$K=1/2 \quad \{\lambda(x) > 1/2\} = \{3 < x < 6\}$$

$$\{\lambda(x) = 1/2\} = \{2 < x < 3\}$$

$$P_0(\lambda(X) > 1/2) + \gamma P_0(\lambda(X) = 1/2) = 0.05$$

$$\underbrace{P_0(3 < X < 6)}_{=0} + \underbrace{\gamma P_0(2 < X < 3)}_{=1/2} = 0.05$$

$$\gamma = 0.01$$

$$\text{MP size-}\alpha \text{ test: } \phi_1(x) = \begin{cases} 1 & \text{if } 3 < x < 6 \\ 0.1 & \text{if } 2 < x < 3 \\ 0 & \text{if } 1 < x < 2 \end{cases}$$

$$\begin{aligned}\text{Power of } \phi &= P_1(3 < X < 6) + 0.1 P_1(2 < X < 3) \\ &= \frac{3}{4} + 0.1 \times \frac{1}{4} = \frac{31}{40}\end{aligned}$$

Consider $\phi_2(x) = \begin{cases} 1 & \text{if } 3 < x < 6 \\ \frac{x-2}{5} & \text{if } 2 < x < 3 \\ 0 & \text{if } 1 < x < 2 \end{cases}$ is also MP size- α test

$$\left(\begin{array}{l} 1. \ E_0 \phi_2 = \alpha = 0.05 \\ 2. \ E_1 \phi_2 = \frac{31}{40} \end{array} \right) \quad (\text{H.W.})$$

Uniqueness:

Result: $X \sim f$. Consider testing $H_0: f = f_0$ vs. $H_1: f = f_1$. For $0 < \alpha < 1$, if a MP size- α test ϕ has power β then $\alpha < \beta$ unless $f_0(x) = f_1(x)$ μ a.e.

Pf: $\alpha \leq \beta$ Consider $\phi_0(x) = \alpha \ \forall x \rightarrow$ size- α test
power of $\phi = \beta \geq$ power of $\phi_0 = \alpha$

if $0 < \alpha = \beta < 1 \Rightarrow \phi_0$ is MP size- α test

$$\Rightarrow \phi_0(x) = \begin{cases} 1 & \text{if } f_1(x) > K f_0(x) \\ 0 & \text{if } < \end{cases}$$

But $\phi_0(x) = \alpha \in (0, 1)$

$$\Rightarrow f_1(x) = K f_0(x) \quad \left(\because \begin{array}{l} f_1 > K f_0 \Rightarrow \phi_0 = 1 \\ f_1 < K f_0 \Rightarrow \phi_0 = 0 \end{array} \right)$$

$$\Rightarrow \int f_1(x) dx = K \int f_0(x) dx$$

$$\Rightarrow K = 1 \Rightarrow f_0(x) = f_1(x) \ \mu \text{ a.e.}$$

HW: Suppose ϕ_0 is a MP size- α test for $H_0: \theta = \theta_0$ vs. $H_1: \theta = \theta_1$ with power $= \beta < 1$. Then $(1 - \phi_0)$ is a MP size- $(1 - \beta)$ test for testing $H_0: \theta = \theta_1$ vs. $H_1: \theta = \theta_0$

HW: Suppose $X_1, \dots, X_n \stackrel{iid}{\sim} f(x|\theta); \theta \in \Theta$
 $H_0: \theta = \theta_0$ vs. $H_1: \theta = \theta_1$

Suppose ϕ_n is a MP size- α test from NP lemma and
for $H_0: \theta = \theta_0$ vs. $H_1: \theta = \theta_1$

$$P_{\theta_0} \left(\frac{f(x|\theta_1)}{f(x|\theta_0)} = K \right) = 0 \quad \forall K \geq 0$$

Show, $\beta_{\phi_n}(\theta_1) \rightarrow 1$ as $n \rightarrow \infty$

UMP Test for simple vs. composite

$$X \sim f(x|\theta): \theta \in \Theta$$

Test: $H_0: \theta = \theta_0$ vs. $H_1: \theta \in \Theta_1$ where $\theta_0 \notin \Theta_1$

1. Consider a MP size- α test ϕ for $H_0: \theta = \theta_0$ vs. $H_1': \theta = \theta_1$
2. If ϕ does not depend on θ_1 , then ϕ is MP size- α $\forall \theta_1 \in \Theta_1$

Hence, ϕ is UMP size- α test for $H_0: \theta = \theta_0$ vs. $H_1: \theta \in \Theta_1$

Ex: $X_1, \dots, X_{10} \sim \text{Ber}(p)$ $H_0: p = \overset{p_0}{0.2}$ vs. $H_1: p \neq 0.2$
for $H_0: \theta = \theta_0$ vs. $H_1: \theta \in \Theta_1$. Find UMP size- α test for $\alpha = 0.05$

Fix $p_1 > 0.2$

$$\frac{f(x|\phi_1)}{f(x|\phi_0)} = \frac{p_1^{\sum x_i} (1-p_1)^{n-\sum x_i}}{p_0^{\sum x_i} (1-p_0)^{n-\sum x_i}} = \left(\frac{p_1}{1-p_1}\right)^{\sum x_i} \left(\frac{1-p_0}{p_0}\right)^{\sum x_i} \left(\frac{1-p_1}{1-p_0}\right)^n$$

$\uparrow \sum x_i$

$$\Leftrightarrow \sum x_i > c$$

Find smallest c : $\mathbb{P}_{p_0}\left(\underbrace{\sum_{i=1}^{10} X_i}_{\text{Bin}(10, 0.2)} > c\right) < 0.05$

$$\phi_1(x) = \begin{cases} 1 & \text{if } \sum x_i > c \\ \gamma & \text{if } = \\ 0 & \text{if } < \end{cases}$$

Solve $\mathbb{P}_{p_0}(\sum X_i > c) + \gamma \mathbb{P}_{p_0}(\sum X_i = c) = 0.05$

HW

$$c = 4$$

Show : γ, c will NOT depend on p_1

Hence ϕ

Ex.

$X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta)$: Test $H_0: \theta = \theta_0$ vs. $H_1: \theta \neq \theta_0$

Show $\phi^*(x) = \begin{cases} 1 & \text{if } x_{(n)} \geq \theta_0 \text{ or } x_{(n)} \leq \theta_0 \alpha^{1/n} \\ 0 & \text{o.w.} \end{cases}$

For $\theta > \theta_0$

We know the UMP size- α test for $H_0: \theta = \theta_0$ vs. $H_1': \theta > \theta_0$ is

$$\phi_1(x) = \begin{cases} 1 & \text{if } x_{(n)} > \theta_0 (1-\alpha)^{1/n} \\ 0 & \leq \end{cases}$$

$$\begin{aligned} \beta_{\phi_1}(\theta) &= \mathbb{P}(X_{(n)} > \theta_0 (1-\alpha)^{1/n}) \\ &= 1 - \left(\frac{\theta_0}{\theta}\right)^n (1-\alpha) \uparrow \theta \\ &\quad \forall \theta > \theta_0 \end{aligned}$$

First, check ϕ^* is a size- α test

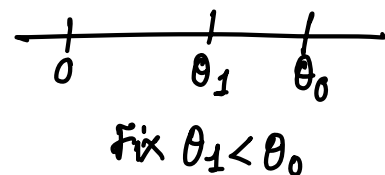
$$\beta_{\phi^*}(\theta) = \mathbb{P}_{\theta}(X_{(n)} \geq \theta_0) + \mathbb{P}_{\theta}(X_{(n)} \leq \theta_0 \alpha^{1/n})$$

$$= 1 - \left(\frac{\theta_0}{\theta}\right)^n + \left(\frac{\theta_0}{\theta}\right)^n \alpha$$

$$= 1 - (1 - \alpha) \left(\frac{\theta_0}{\theta}\right)^n \quad \forall \theta > \theta_0$$

For $\theta < \theta_0$:

Test $H_0: \theta = \theta_0$ vs. $H_1: \theta < \theta_0$



$$\phi_2(\alpha) = \begin{cases} 1 & \text{if } X_{(n)} < c \\ 0 & \text{if } X_{(n)} > c \end{cases} \quad \text{s.t. } \mathbb{P}_{\theta_0}(X_{(n)} < c) = \alpha$$

$$\Rightarrow \left(\frac{c}{\theta_0}\right)^n = \alpha$$

$$\Rightarrow c = \theta_0 \alpha^{1/n}$$

$\rightarrow$ is a UMP size- α test for $\theta = \theta_0$ vs. $\theta < \theta_0$

$$\beta_{\phi_2}(\theta) = \mathbb{P}_{\theta}(X_{(n)} < \theta_0 \alpha^{1/n}) = \left(\frac{\theta_0}{\theta}\right)^n \alpha \downarrow \text{ for } \theta < \theta_0$$

For $\theta < \theta_0$:

$$\beta_{\phi^*}(\theta) = \begin{cases} \left(\frac{\theta_0}{\theta}\right)^n \alpha & \text{if } \theta > \alpha^{1/n} \theta_0 \\ 1 & \text{if } \theta < \alpha^{1/n} \theta_0 \end{cases}$$

$$\therefore \beta_{\phi^*}(\theta) \geq \beta_{\phi_2}(\theta) \quad \forall \theta < \theta_0$$